

Master Thesis
BlueRov2 Formation Control

Week 6

João Zenário

Next Steps

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents

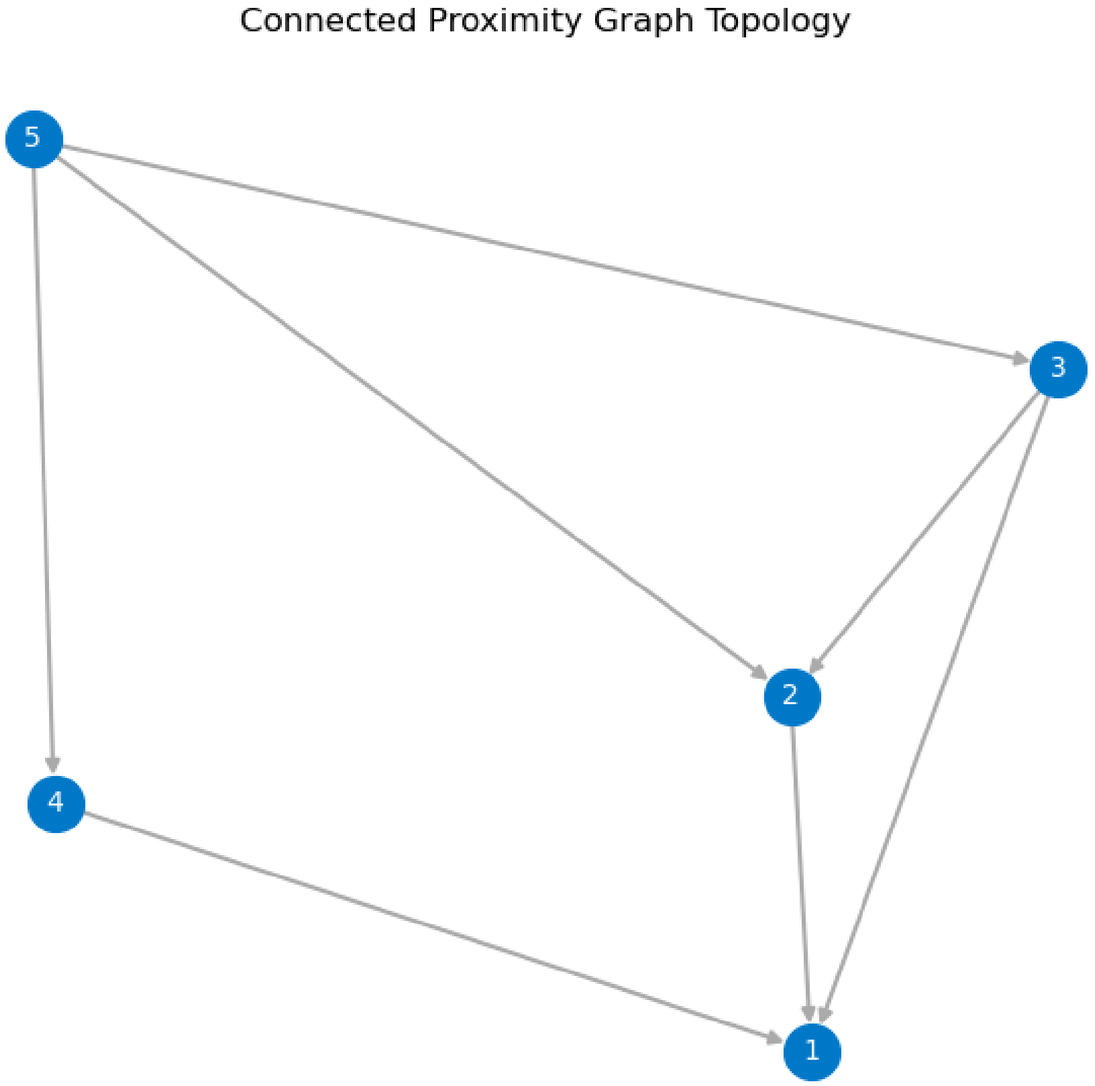


- BlueRov2 model no input constraints
  -
- [Research challenge] integrate input constraints
  -
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Achievements

- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis
  - Khalil's book on Nonlinear Control

- Implementation
  - Improvements
    - Generalized and modularized the code
    - Added Formation Offset error
    - Tuned and changed variables into more realistic ones
  - Implemented Field of View constraint


Connected Proximity Graph Topology
5
3
2
4
1

Field of View Constraint

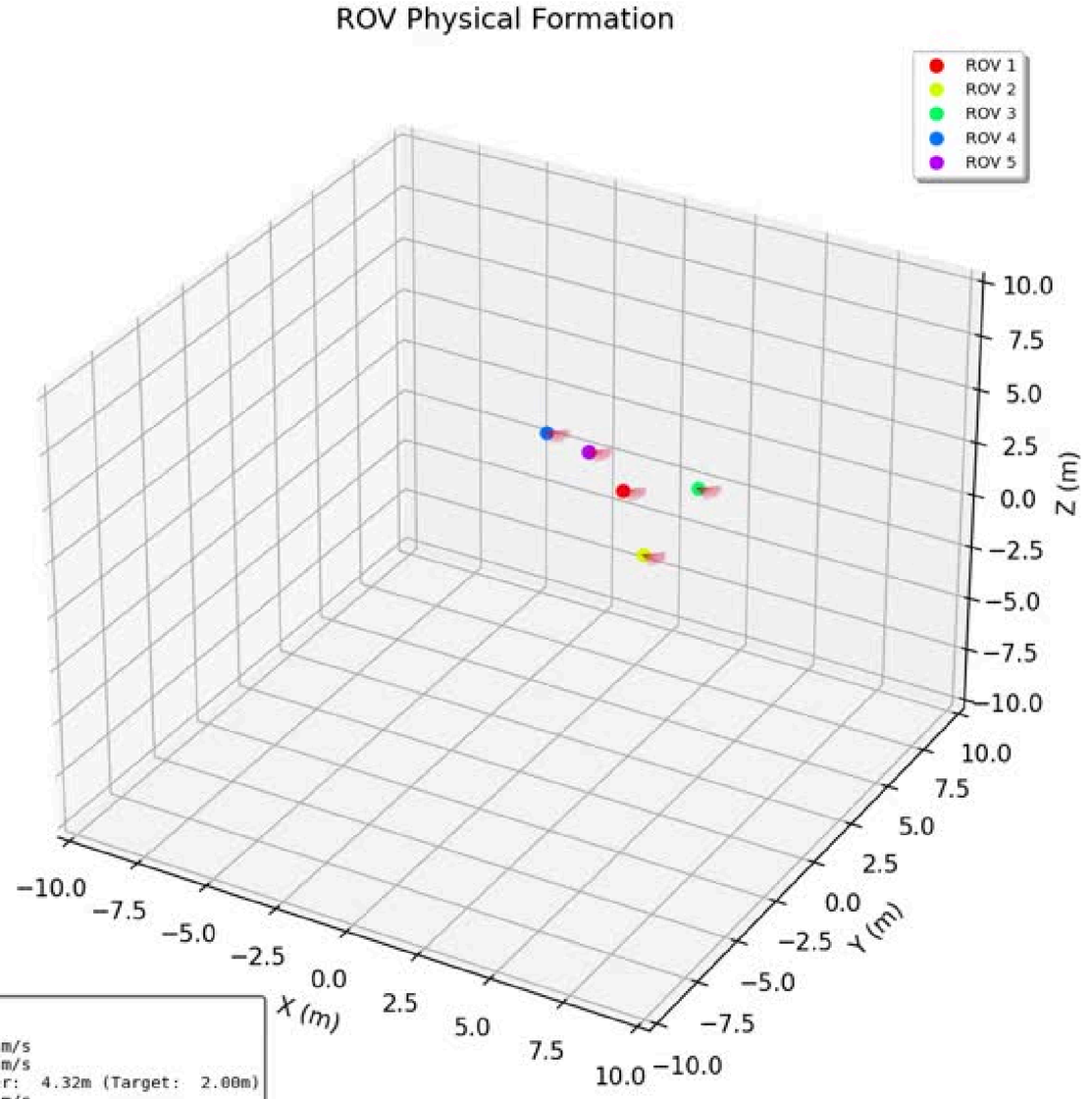
Barrier Function

$$W_{\alpha,k}(\alpha_k) = \frac{1}{2}\left[\alpha_k^2 + \ln\left(\frac{\Delta_{\alpha,k}^2}{\Delta_{\alpha,k}^2 - \alpha_k^2}\right)\right]$$

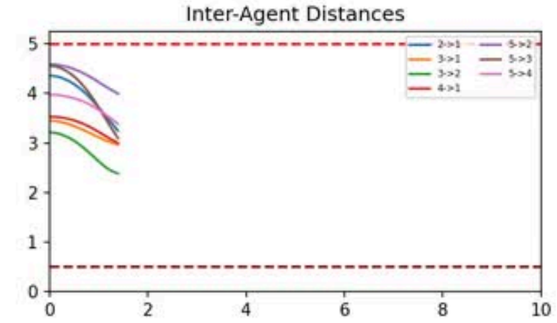
$$\nabla W(\alpha_k) = \left(\frac{1}{\Delta_{\alpha,k}^2 - \alpha_k^2}\right)\alpha_k$$

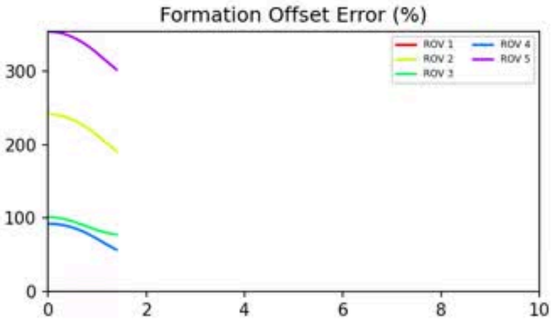
$$\omega_i = -k_p \sum_{j\in\mathcal{N}_i} \alpha_k - c_3 \sum_{k\in\mathcal{N}_i} \nabla W(\alpha_k)$$

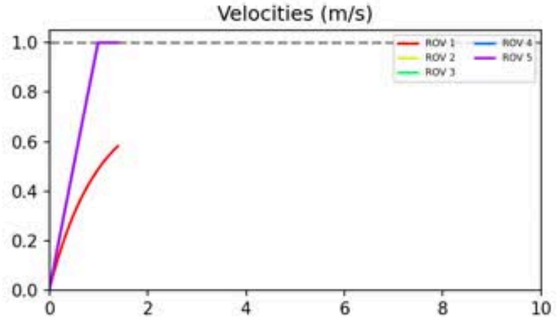
$$\alpha_k = \psi_i - \psi_j$$

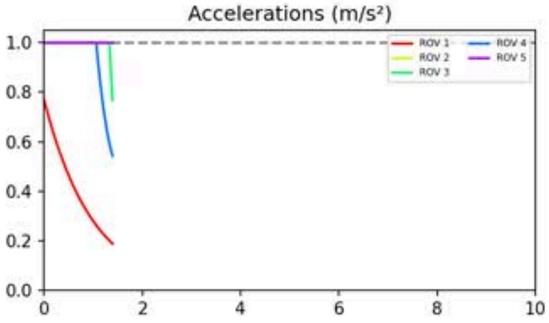


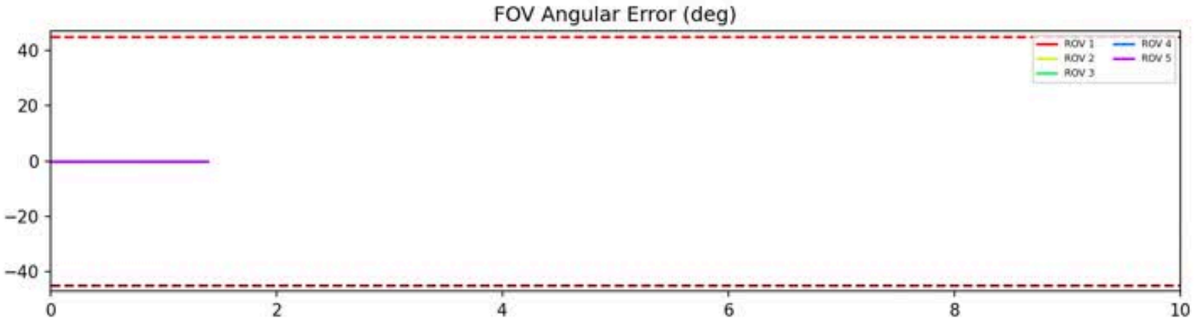

```
Time:  0.21s
---------------
ROV 1 Vel:  0.16 m/s
ROV 2 Vel:  0.23 m/s
Distance to Leader:  4.32m (Target:  2.00m)
ROV 3 Vel:  0.23 m/s
Distance to Leader:  3.42m (Target:  2.00m)
ROV 4 Vel:  0.23 m/s
Distance to Leader:  3.51m (Target:  2.00m)
ROV 5 Vel:  0.23 m/s
Distance to Leader:  5.49m (Target:  2.00m)
```











Field of View Constraint

Barrier Function

$$W_{\alpha,k}(\alpha_k) = \frac{1}{2}\left[\alpha_k^2 + \ln\left(\frac{\Delta_{\alpha,k}^2}{\Delta_{\alpha,k}^2 - \alpha_k^2}\right)\right]$$

$$\nabla W(\alpha_k) = \left(\frac{1}{\Delta_{\alpha,k}^2 - \alpha_k^2}\right)\alpha_k$$

$$\omega_i = -k_p \sum_{j \in \mathcal{N}_i} \alpha_k - c_3 \sum_{k \in \mathcal{N}_i} \nabla W(\alpha_k)$$

$$\alpha_k = \psi_i - \psi_j$$

Field of View Constraint

Barrier Function

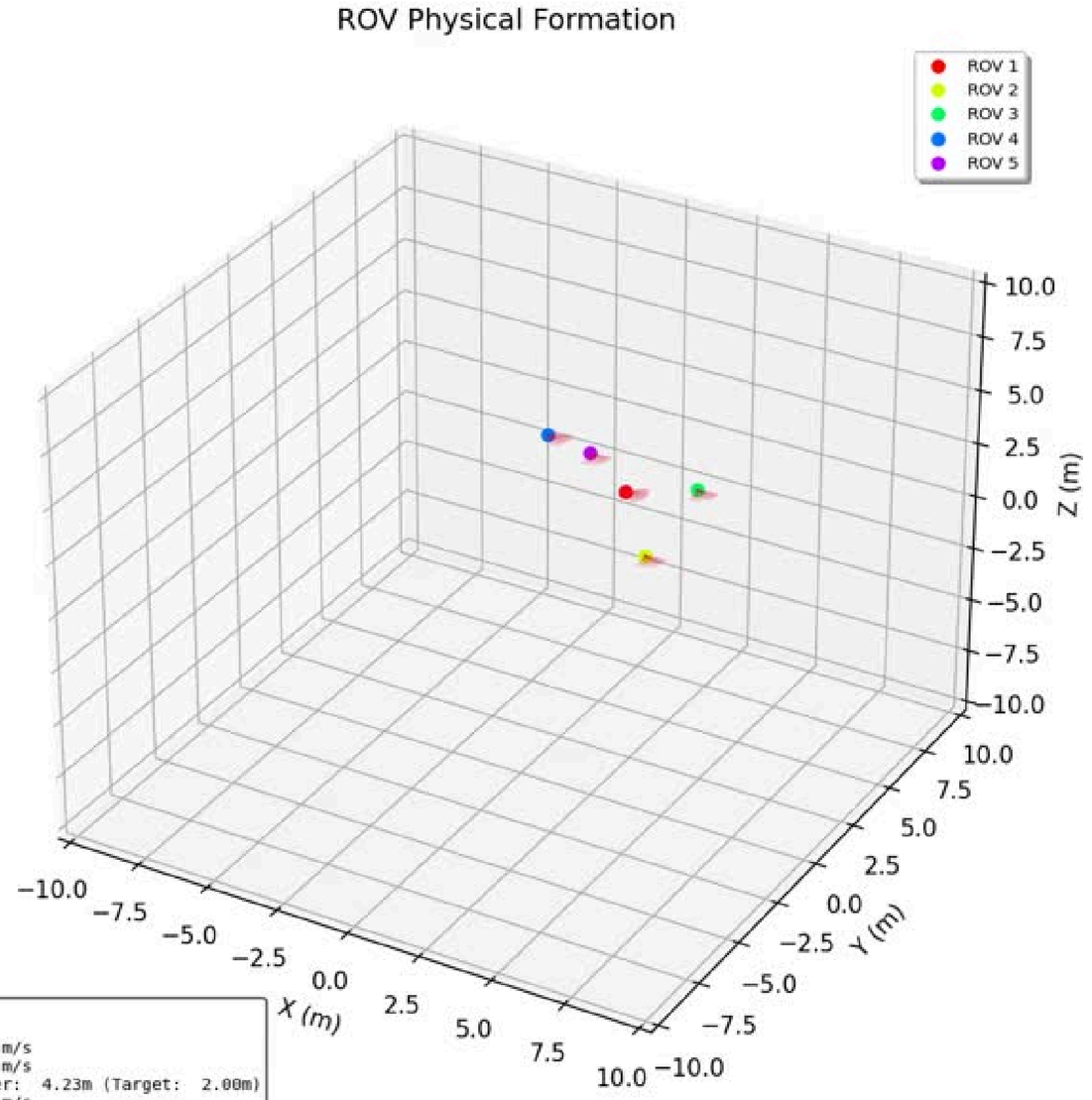
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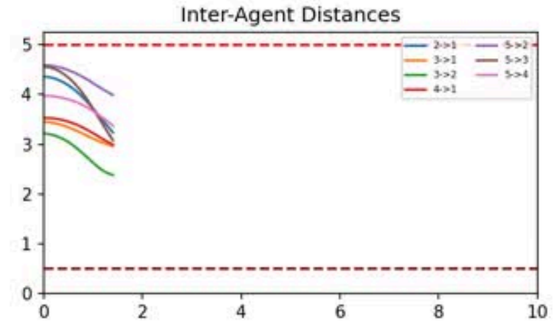
$$\omega_i = -k_p \sum_{j \in \mathcal{N}_i} \alpha_k - c_3 \sum_{k \in \mathcal{N}_i} \nabla W(\alpha_k)$$

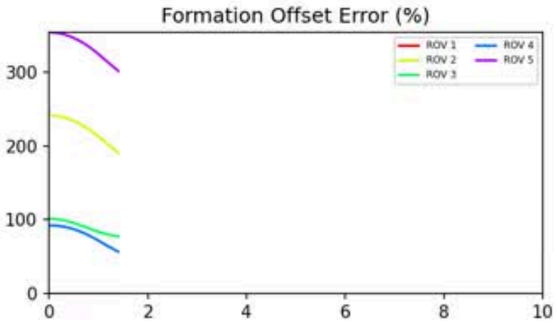
$$\alpha_k = \psi_i - \psi_{target}$$

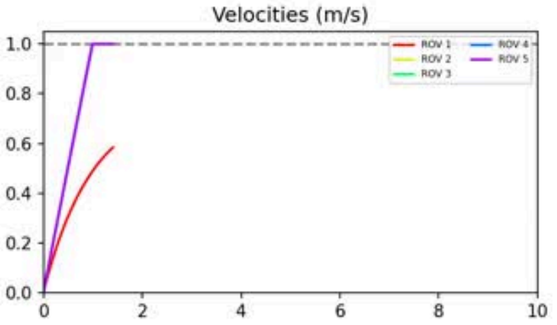
$$\omega_i = -k_p \alpha_k - c_3 \nabla W(\alpha_k)$$

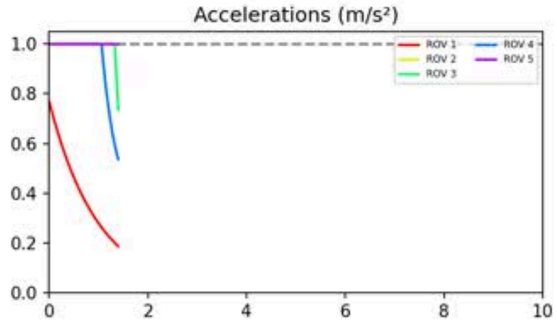


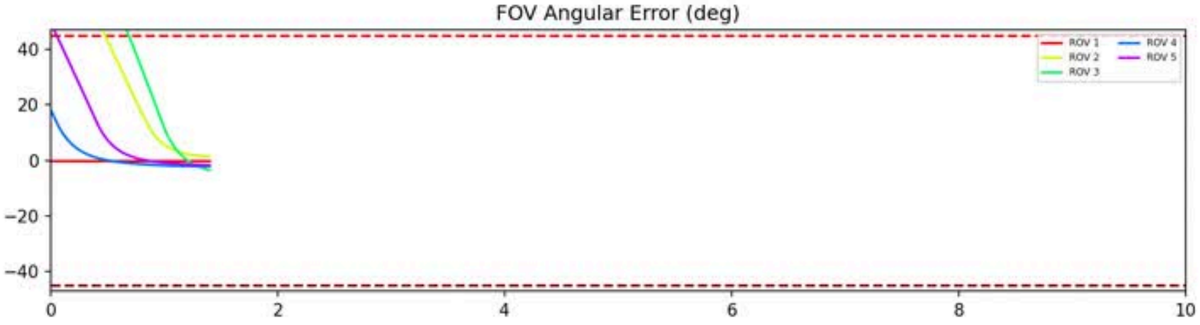

```
Time:  0.42s
---------------
ROV 1 Vel:  0.27 m/s
ROV 2 Vel:  0.44 m/s
Distance to Leader:  4.23m (Target:  2.00m)
ROV 3 Vel:  0.44 m/s
Distance to Leader:  3.37m (Target:  2.00m)
ROV 4 Vel:  0.44 m/s
Distance to Leader:  3.47m (Target:  2.00m)
ROV 5 Vel:  0.44 m/s
Distance to Leader:  5.42m (Target:  2.00m)
```











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Questions & Comments